Soohyun Cha (차수현)

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WORK EXPERIENCES

Samsung Research, Samsung Electronics — Seoul, Republic of Korea

AI/Software Research Engineer at the AI Center — Mar. 2026 – **Current**

- (2026.07~) Efficient serving of LLMs on edge GPUs

Undergraduate Internship at the Communication Research Center — July 2022 – Aug. 2022

- Designing a deep learning-based KPI predictor for communication systems

EDUCATION

Seoul National University — Seoul, Republic of Korea

Master of Science in Electrical and Computer Engineering — Mar. 2024 – Feb. 2026

- Advisor: Prof. Jaewoong Sim
- Research interest: Computer systems and algorithms for efficient serving of emerging machine learning workloads, such as large language models (LLMs)
- Thesis: Efficient Serving of Large Language Models with Adaptive Speculative Decoding
- GPA: 4.04 / 4.30

Bachelor of Science in Electrical and Computer Engineering — Mar. 2018 – Feb. 2024

- GPA: 3.74 / 4.30, Major GPA: 3.80 / 4.30 (Graduated with *Cum Laude*)
- Full-tuition scholarship for 8 semesters by the National Science and Engineering Scholarship, Korea Student Aid Foundation (KOSAF)

PUBLICATIONS

MX+: Pushing the Limits of Microscaling Formats for Efficient Large Language Model Serving

Jungi Lee, Junyong Park, **Soohyun Cha**, Jaehoon Cho, Jaewoong Sim

Proc. of the 58th International Symposium on Microarchitecture (MICRO), Seoul, Korea, Oct 2025

TEACHING EXPERIENCE

Graduate Teaching Assistant at Seoul National University

- Digital Systems Design and Experiments, ECE 315.A — Fall 2024
  - Lab session for Verilog and FPGA board practices
  - Q&A session for implementing end-to-end CNN accelerator on FPGA board
- Computer Organization, ECE 322 — Spring 2024
  - Q&A session for implementing a pipelined CPU with a branch predictor in RTL and a cycle-level simulator for a cache system in C++
  - Recitation session for reviewing topics on CPU microarchitecture

Undergraduate Course Tutor at Seoul National University

- Digital Systems Design and Experiments, ECE 315.A — Fall 2023
  - Q&A for Verilog, FSM (finite state machine), and digital arithmetic/numerics
- Basic Computing, Faculty of Liberal Education — Fall/Spring 2023, Fall/Spring 2022

- Q&A for Python tutorial covering Numpy, Matplotlib, and Pandas
- Outstanding tutor award (Fall 2023)

SKILLS

Programming Language: Python, Verilog/SystemVerilog, C/C++, CUDA
Frameworks: PyTorch, Hugging Face Transformers, Triton, vLLM
Tools: Nsight Systems, Xilinx Vivado
Platforms: Linux, Artix-7 FPGA board
Foreign Language: English (TOEIC Speaking: Intermediate High)

MISCELLANEOUS

Military Service at the Republic of Korea Army July 2020 – Jan. 2022

Completed 18 months of mandatory military service in the Republic of Korea Army